

# Yu Li

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## Education

**George Washington University** **Washington, D.C.**  
*Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0*    Aug 2025 – May 2029  
 Advisor: [Prof. Tian Lan](#) 🌐; Mentor: [Prof. Zhengling Qi](#) 🌐 (expected)

**Wuhan University, Hongyi Honor College** **Wuhan, China**  
*B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0*    Sept 2021 – Jun 2025  
 Advisor: [Prof. Cheng Lei](#) 🌐 & [Prof. Wei Liu](#) 🌐

## Research Interests

My research focuses on **LLM post-training**, **agent policy learning**, and **generative AI**. Specific topics include LoRA fine-tuning, preference optimization, reasoning, agentic RL, and multimodal in-context learning.

## Industry Experience

**ByteDance Inc.** **San Jose, CA**  
*Machine Learning Scientist Intern, Tiktok Recommendation Core*    May 2026 – Aug 2026  
 Worked on ML-driven video recommendation for TikTok’s production system, focusing on content understanding and LLM-based ranking within a cross-functional team.

## Preprints & Under Review

- [P.1] [OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning](#) 🌐  
 Yu Li, Rui Miao, Tian Lan, Zhengling Qi
- [P.2] [ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning](#) 🌐  
 Yu Li, Rui Miao, Zhengling Qi, Tian Lan
- [P.3] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**  
 Yu Li, Binxu Li, Tian Lan  
*Under review at COLM, Avg rating:6.5*
- [P.4] [Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO](#) 🌐  
 Yu Li, Tian Lan, Zhengling Qi
- [P.5] [INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization](#) 🌐  
 Yu Li, Tian Lan, Zhengling Qi  
 AI with Recursive Self-Improvement Workshop at ICLR, 2026

## Publications

C=Conference, J=Journal, †=Equal Contribution

- [C.1] **CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion**  
 Yu Li, Yujun Cai, Chi Zhang  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026*

**[C.2] Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization**

Yu Li, Sizhe Tang, Tian Lan

*Findings of the Association for Computational Linguistics (ACL), 2026*

**[C.3] KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models**

Yu Li, Da Chang, Xi Xiao

*IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) [Oral], 2026*

**[C.4] ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense**

Yu Li†, Sizhe Tang†, Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan

*Intelligent Cloud Computing and Networking Workshop at INFOCOM, 2026*

**[C.5] Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT**

Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang

*The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026*

**[C.6] Mixed Text Recognition with Efficient Parameter Fine-Tuning and Transformer**

Yu Li†, Da Chang†

*International Conference on Neural Information Processing (ICONIP), 2024*

**[J.1] Dual branch SAM-Transformer Fusion Network for Accurate Breast Ultrasound Image Segmentation**

Yu Li, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

*Medical Physics, 2025*

**[J.2] SfMDiffusion: Self-supervised Monocular Depth Estimation in Endoscopy Based on Diffusion Models**

Yu Li, Da Chang, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

*International Journal of Computer Assisted Radiology and Surgery, 2025*

**[J.3] Local Optimum Time-Reassigned Synchrosqueezing Transform for Bearing Fault Diagnosis**

Site Lv, Shan Zeng, Yu Li, Ke Yang, Yulong Chen

*IEEE Sensors Journal, 2024*

## Selected Awards & Honors

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- **ICML 2026 Gold Reviewer**, top reviewers recognition. 2026
  - **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. 2024
  - **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. 2023, 2024
  - **First-Class Scholarship**, Top 5%, ¥3,000, three times. 2022, 2023, 2024

## Academic Services

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- **Conference Reviewer:** {NeurIPS, COLM, ACM MM, ICML, ICLR, AAAI, ICASSP}'2026
  - **Journal Reviewer:** TMLR, TPAMI, ToN, Neurocomputing

## Skills

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- **Languages:** English (TOEFL 105), Chinese (Native), Japanese (Basic)
  - **ML/LLM Stack:** PyTorch, DeepSpeed, Flash Attention, vLLM, OpenRLHF, VERL, TRL
  - **Tools & Platforms:** Linux/Ubuntu, Docker, Git, Weights & Biases, Cadence